

ESG Intelligence Briefing — July 12, 2026

Executive Summary

Kazakhstan's construction and export sectors face immediate challenges and opportunities. The EU's CBAM, effective this year, poses significant carbon tax liabilities, particularly for steel and aluminum exporters, with potential costs exceeding 123% of some steel export values if actual emissions are not reported. Simultaneously, a new Kazakhstan Construction Code came into effect on July 1, 2026, strengthening control but lacking explicit green building mandates, while global trends show a strong push towards low-carbon concrete alternatives driven by procurement rules and material innovation. Construction clients must urgently assess and report embodied carbon, explore low-carbon material substitutions, and engage with evolving domestic standards.

What Changed Since Last Briefing

This is the first briefing generated by the ESG Intelligence Agent, establishing a baseline for future comparisons.

1. Regulation Watch

- **EU Carbon Border Adjustment Mechanism (CBAM) Impact on Kazakhstan**
 - **What it is:** The EU's CBAM aims to equalize carbon costs between EU and imported goods, impacting products like steel, iron, and aluminum. While the transitional phase focused on reporting, the financial implications are now becoming concrete.
 - **Who it affects:** Directly impacts Kazakhstani exporters of steel, iron, and aluminum to the EU market. Construction clients sourcing materials internationally or involved in export supply chains will be affected.
 - **Effective date:** Reporting requirements were established, and financial obligations are now materializing, with costs based on EU carbon prices (around €80/ton).
 - **Business implication:** Failure to accurately measure, calculate, and report actual emissions will result in punitive default values, which are significantly higher. The International Trade Centre (ITC) estimates potential CBAM costs for Kazakhstan's steel exports could exceed 123% of their value if default values are applied, compared to a potentially halved liability (from €114 million to €57 million annually) with accurate reporting. Kazakhstan's low domestic carbon price (€0.40/ton) means exporters effectively pay the full EU carbon price.
 - **Source:** [Kazakhstan's Exporters Race to Adapt to Europe's Carbon Border Tax - The Astana Times (June 4, 2026)](<https://astanatimes.com/2026/06/kazakhstans-exporters-race-to-adapt-to-europes-carbon-border-tax>)
- **Kazakhstan Construction Code Effective July 1, 2026**
 - **What it is:** A new comprehensive Construction Code, signed January 9, 2026, replaces previous laws and streamlines regulatory provisions for architectural, urban planning, and construction activities.
 - **Who it affects:** All construction developers, contractors, and specialists operating within Kazakhstan.
 - **Effective date:** July 1, 2026.
 - **Business implication:** The Code strengthens state control, introduces a National Institute for Technical Standardization in Construction, and mandates an automated digital urban planning cadaster. While focusing on safety and proper development, the reviewed document does not explicitly detail new green building standards or specific sustainability mandates. However, the establishment of a standardization institute suggests future potential for integrating environmental performance metrics. It also sets statutory guarantee periods of no less than 10 years for supporting structures and 5 years for other works, influencing material

durability considerations.

- **Source:** [Kazakhstan Construction Code Goes into Effect - Morgan Lewis (June 30, 2026)](<https://www.morganlewis.com/pubs/2026/06/kazakhstan-construction-code-goes-into-effect>)

2. Construction & Market News

- **Global Shift to Low-Carbon Concrete Accelerates**
- **What it is:** The construction industry is rapidly adopting low-carbon concrete solutions, moving beyond pilot projects to widespread application in infrastructure, commercial, and residential sectors. This is driven by both climate goals and evolving procurement requirements.
- **Key developments:** Portland-limestone cement (PLC), offering approximately a 10% reduction in embodied carbon compared to ordinary Portland cement, is becoming a practical first step due to its ease of integration. Limestone calcined clay cement (LC3) is gaining significant traction, with potential to reduce carbon intensity by up to 40% due to the wider availability of clay compared to other supplementary cementitious materials like fly ash or slag. New cement chemistries and cement-free binders are also emerging from the lab to commercial projects.
- **Source:** [Low-Carbon Concrete: How Cement Alternatives Are Changing Construction In 2026 - Carbo Europe](<https://www.carboeurope.org/low-carbon-concrete>)

3. ESG Disclosure Signals

- **Procurement Rules Driving Embodied Carbon Disclosure**
- **What it is:** Global government bodies, such as the U.S. General Services Administration (GSA), are implementing procurement rules that set Global Warming Potential (GWP) limits for concrete by strength class and require product-specific Type III Environmental Product Declarations (EPDs).
- **Signal:** This signals a market-wide shift where documented low-carbon performance is becoming a prerequisite for securing public and, increasingly, private contracts. Companies failing to provide verified embodied carbon data (via EPDs) risk losing significant project opportunities. This trend will likely expand globally and to other material categories.
- **Source:** [Low-Carbon Concrete: How Cement Alternatives Are Changing Construction In 2026 - Carbo Europe](<https://www.carboeurope.org/low-carbon-concrete>)

4. Embodied Carbon Spotlight

- **Note:** *Due to a technical issue with the embodied carbon lookup tool, specific quantitative LCA results for a hypothetical scenario cannot be provided in this briefing. The following discussion of hotspots and circular economy recommendations is based on general industry knowledge and the trends identified in the "Low-Carbon Concrete" article.*
- **Scenario Focus (Qualitative):** A typical concrete-frame building structure, considering the high impact of cement production and the relevance of steel to CBAM.
- **Key Embodied Carbon Hotspots:**
- **Cement Production:** Remains the single largest contributor to concrete's embodied carbon due to both the energy-intensive clinker production and the chemical process of calcination.
- **Steel Production:** Especially primary steel production, is highly carbon-intensive. This is directly relevant to Kazakhstan's export challenges under CBAM.
- **Circular Economy Recommendations (Qualitative):**

Lower-carbon switch	Saving (Indicative)	Rationale
Concrete: Replace Ordinary Portland Cement (OPC) with Portland-Limestone Cement (PLC)	~10% reduction in concrete's GWP	PLC is a direct, performance-comparable substitute with lower clinker content, widely adopted.
Concrete: Utilize Limestone Calcined Clay Cement (LC3) or other Supplementary Cementitious Materials (SCMs)	Up to 40% reduction in concrete's GWP for LC3	LC3 uses abundant calcined clay, offering significant carbon cuts. SCMs like fly ash/slag substitute clinker.
Steel: Specify Electric Arc Furnace (EAF) steel from recycled scrap	Up to 70-80% reduction vs. virgin BF-BOF steel	EAF using high scrap content drastically cuts primary production emissions, highly relevant for CBAM.
Design: Optimize structural design for material efficiency	Varies significantly by project, ~5-15%	Lean design, prefabrication, and modular construction reduce overall material consumption.
Material Reuse/Recycling: Integrate recycled aggregates in concrete, reuse steel components	Varies, can be substantial	Reduces demand for virgin materials and avoids end-of-life emissions.

Embodied Carbon Trends

The market is rapidly moving towards verified low-carbon material solutions. The read article highlights that:

- **EPDs are critical:** Environmental Product Declarations (EPDs) are becoming standard requirements, particularly in public procurement, forcing manufacturers to quantify and disclose their product's embodied carbon.
- **Innovation in Binders:** Beyond traditional SCMs, new cement chemistries and cement-free binders are progressing from R&D to commercial application, promising even deeper decarbonization.
- **Demand-side drivers:** Regulations and procurement policies are now significant drivers, shifting the industry from voluntary adoption to mandated low-carbon material use, pushing down carbon factors for common materials.

Recommended Actions

1. **Prioritize CBAM Compliance & Data Accuracy:** For any client involved in exporting steel, aluminum, or other CBAM-covered materials to the EU, immediately invest in systems for precise measurement, verification, and reporting of actual production emissions. This is critical to avoid significantly higher default carbon charges and maintain market access.
2. **Evaluate Low-Carbon Concrete Specifications:** Review current concrete specifications for all projects. Actively explore and specify Portland-Limestone Cement (PLC) as a baseline low-carbon option. Investigate the feasibility and availability of Limestone Calcined Clay Cement (LC3) or other advanced SCMs for deeper decarbonization, especially for large-volume pours.
3. **Demand Verified Carbon Data (EPDs):** Begin requiring product-specific Type III Environmental Product Declarations (EPDs) from all material suppliers, especially for high-impact materials like concrete and steel. This will be essential for meeting future procurement requirements and accurately tracking project-level embodied carbon.

4. Engage with National Standardization Bodies: Proactively monitor the development of the new National Institute of Technical Standardization in Construction in Kazakhstan. Advocate for the inclusion of green building standards and embodied carbon metrics within future technical regulations to align with international best practices and prepare for future domestic requirements.

5. Explore EAF Steel Sourcing: For projects requiring steel, assess the embodied carbon of available options and prioritize sourcing Electric Arc Furnace (EAF) steel with high recycled content to significantly reduce embodied carbon and mitigate CBAM exposure where applicable.

Sources

- [Kazakhstan's Exporters Race to Adapt to Europe's Carbon Border Tax - The Astana Times](<https://astanatimes.com/2026/06/kazakhstans-exporters-race-to-adapt-to-europes-carbon-border-tax>)
- [Kazakhstan Construction Code Goes into Effect - Morgan Lewis](<https://www.morganlewis.com/pubs/2026/06/kazakhstan-construction-code-goes-into-effect>)
- [Low-Carbon Concrete: How Cement Alternatives Are Changing Construction In 2026 - Carbo Europe](<https://www.carboeurope.org/low-carbon-concrete>)

Generated by the ESG Intelligence Agent on July 12, 2026. Embodied-carbon figures are indicative cradle-to-gate (A1-A3) estimates — verify with product-specific EPDs before formal reporting.